

Zinuo (Henry) You

GENERATIVE MODELLING & WORLD MODELS · RL & DISTILLATION · OPTIMIZATION

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PROFILE

PhD researcher in Engineering Mathematics and Computer Science at the University of Bristol, working at the intersection of generative modelling, reinforcement learning, and optimization. Research spans long-horizon consistency and parameter-efficient post-training for video/world models; diffusion, flow matching, and Schrödinger bridges; and efficient generative inference. First-author publications include an ICML 2026 Spotlight, AAAI 2026, and ICASSP 2024. Experienced across problem formulation, implementation, multi-GPU training, and cross-task evaluation.

EDUCATION

University of Bristol · PhD, Engineering Mathematics and Computer Science · 2023-present

University of Sheffield · MSc, Electronic and Electrical Engineering, Distinction · 2019-2020

Southwest University · Bachelor's, Electronic Science and Technology / Solid-State Physics · 2015-2019

TECHNICAL SKILLS

Generative models: DiT/MMDiT, latent diffusion, flow matching, T2V/I2V/Action2V, image generation, video restoration, long-horizon rollout.

RL & optimization: offline/online RL, pathwise policy optimization, reward/preference fine-tuning, finite-operator distillation, Bayesian optimization.

Systems: Python/PyTorch, LoRA/PEFT, SFT/DPO/PPO/GRPO, ODEs/SDEs, low-NFE sampling, Slurm/HPC, vLLM, INT8 quantization/NPU.

CORE RESEARCH

Finite-Time Corrective Transport for Unified Optimization and Distillation

Working paper · 2025-present

- Learn an NFE-conditioned corrective field over a frozen base transport, directly optimizing the deployed finite-step operator while keeping the generator and solver rule fixed.
- Connect critic-free pathwise policy optimization through rollout adjoints/BPTT with conditional kernel score/MMD distribution matching, without dense teacher trajectories.
- Evaluate conditional generation and restoration using I²SB, PMRF, SeedVR, SANA, and UNSB under matched NFE and optimization/reward-query budgets. Gains over corresponding baselines are approximately 11%-32% across the evaluated metrics.

Latent Laplace Diffusion for Irregular Multivariate Time Series

ICML 2026 Spotlight · 2023–2025

- Learn conditional diffusion in VAE latent space and generate complete forecast windows in parallel at query timestamps, avoiding regriding and sequential physical-time integration.
- Parameterize stable trajectories with complex-conjugate poles and damping constraints, with renewal averaging and interval-aware history encoding.
- Across seven datasets and 2%-92% missingness, report 1.2×-130× faster inference and 11%-29% lower CRPS/MSE/MAE under matched protocols.

INDUSTRY RESEARCH

Long-Horizon Consistency for Causal Video & Interactive World Models

Samsung AI Center-Cambridge · Research Intern · Mar-Sep 2026

- Build Wan2.1-1.3B + Causal-rCM T2V/Action2V evaluation across 5, 10, 20, and 40+ chunks; measure consistency and action adherence with VBench-Long/I2V and WBench/WorldRoamBench.
- Probe VAE latents, DiT/MMDiT hidden states, and KV histories; compare online, refreshed, and recomputed strategies across HunyuanVideo-1.5-8B + minWM and Wan2.2-5B + BiWM.
- Short-horizon drift predicts long-horizon failure (Spearman $\rho = 0.61-0.70$). Partition-consistency LoRA on 2×H100 reduces rollout inconsistency/drift by approximately 25%/20%, improving VBench-Long/WBench by 3%-8%.

ADDITIONAL RESEARCH & EXPERIENCE

Automotive Engineer Intern

Qualcomm · Apr-Nov 2021

- Support INT8 NPU deployment of real-time automotive object and trajectory detection through FP32-INT8 alignment, operator-compatibility analysis, error localization, and end-to-end evaluation.
- Achieve approximately 40 FPS in steady-state on-device tests; establish benchmarks and failure attribution across quantization, module outputs, and runtime performance.

GNN Architecture Capacity and Financial Graph Learning

AAAI 2026 / ICASSP 2024 / ICAART 2024 · 2023-2025

- Develop channel-capacity-constrained estimation for GNN width and depth, reducing architecture-search time by approximately 10× while improving accuracy/F1 by 2%-11%.
- Design dynamic stock-graph models using relational retention, graph diffusion, and decoupled representations.

Restricted-Domain LLM Audit Agents

EPSRC · Student Research Associate · Feb-Oct 2026

- Build an evidence-constrained agent over millions of UK corporate filings, covering 20 task categories; generate approximately 20,000 SFT examples and 20,000 preference pairs.
- Run parameter-efficient SFT/DPO on 30B-class open models using up to 2×H100, with programmatic evaluation and vLLM batched assessment. Report 81% task pass, 79% correct refusal, and 76% correct evidence citation.

TPE-AS: Adaptive Bayesian Optimization for Black-Box Portfolio Models

UKRI-STRATIPHY · Student Research Associate · Jun 2024-Apr 2025

- Optimize private portfolio models with at least 25 mixed variables and expensive noisy evaluations, using a controlled evaluator isolated from private models, data, and backtesting infrastructure.
- Combine a budget-dependent mean-variance penalty with clipped importance correction. Under 500-evaluation budgets, attain the highest annualized Sharpe in 8/12 scenarios and lowest optimization-trajectory variance in 12/12.

PUBLICATIONS

1. Zinuo You, Jin Zheng, John Cartlidge. [Latent Laplace Diffusion for Irregular Multivariate Time Series](#). ICML 2026. **Spotlight**.
2. Zinuo You, Jin Zheng, John Cartlidge. [How Wide and How Deep? Mitigating Over-squashing of GNNs via Channel Capacity Constrained Estimation](#). AAAI 2026.
3. Zinuo You, Pengju Zhang, Jin Zheng, John Cartlidge. [Multi-Relational Graph Diffusion Neural Network with Parallel Retention for Stock Trends Classification](#). ICASSP 2024.
4. Zinuo You, John Cartlidge, Karen Elliott, Menghan Ge, Daniel Gold. [Improving Bayesian Optimization for Portfolio Management with an Adaptive Scheduling](#). ICAAI 2025.
5. Zinuo You, Zijian Shi, Hongbo Bo, John Cartlidge, Li Zhang, Yan Ge. [DGDNN: Decoupled Graph Diffusion Neural Network for Stock Movement Prediction](#). ICAART 2024.
6. Zinuo You, Zijian Shi, Hongbo Bo, John Cartlidge, Li Zhang, Yan Ge. [Multi-Stock Movements Classification: A Decoupled Graph Diffusion Neural Network Approach](#). LNCS 2025 · ICAART 2024 revised selected papers.

HONORS & FUNDING

University of Bristol Scholarship (2025) · ECEF Fellowship Award (2024)

Jean Golding Institute Seed Corn Fund (2023) · MCM/ICM Honorable Mention (2019)